

ECE 135 Final Report

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Abstract—Anti-resonant hollow core fibers (AR-HCF) are an emerging technology that leverages the anti-resonant properties of their walls to improve the signal-to-noise ratio of traditional HCFs. These cables lacked an analytic model, slowing their commercial adoption. Zeisberger and Schmidt’s paper on “Analytic model for the complex effective index of the leaky modes of tube-type anti-resonant hollow core fibers” derived an analytic model for this exact purpose [1].

With this approach, dispersion, attenuation, and confinement loss could be easily analyzed, reducing guesswork in the manufacturing of AR-HCFs. Utilizing VPI Photonics and Python, the clear advantages of AR-HCFs versus traditional silica fiber are demonstrated.

Silica fibers, compared to AR-HCFs, are limited by power and resonance peaks due to manufacturing constraints. In contrast, AR-HCFs are limited by predetermined resonance peaks and manufacturing that has yet to be scaled. Using these findings, conclusions were drawn about the application and implementation constraints of AR-HCFs.

I. INTRODUCTION

The fundamental limits of silica have led to capped performance of traditional fiber-optic cables. With a higher index of refraction than air ($n_{\text{silica}} = 1.45$), we see a decreased propagation velocity with velocity inversely proportional to the index of refraction, adding a notable latency compared to free space communications:

$$v = \frac{c}{n} \quad (1)$$

Silica as a medium also exhibits a maximum-intensity transmission due to induced scattering, specifically Brillouin, Raman, and Kerr nonlinearities. In addition to scattering from the crystal structure, absorption bands from water introduced during manufacturing create unusable wavelength regions known as OH resonance bands [2].

AR-HCFs operate by guiding light in either air or vacuum, both of which have an index of refraction of approximately $n \approx 1$, resulting in near-free-space communication speeds with fiber-level signal-to-noise ratio (SNR). Without a solid medium, scattering effects are significantly reduced, and there are no intrinsic material absorption bands.

The lack of material enables much higher power input, increasing the potential bit rate through tighter channel spacing, higher launch power, reduced amplifier requirements, and higher SNR at the receiver.

With light interacting with the entire cross-section of silica in traditional fiber, light only reacts with a thin cross-sectional washer in AR-HCFs, leading to a nonlinear overlap factor, η , approximately three orders of magnitude smaller.

$$\eta = \frac{P_{\text{glass}}}{P_{\text{total}}} \approx 10^{-3} \quad (2)$$

$$n_{2,\text{eff}}^{\text{AR-HCF}} = \eta \cdot n_2^{\text{silica}} \quad (3)$$

where P_{glass} is the optical power propagating in the glass cladding and P_{total} is the total guided power. For well-designed AR-HCFs, $\eta \approx 10^{-3}$, meaning less than 0.1% of the light interacts with the glass. Nonlinear effects persist; however, they can now be controlled by design changes, such as thinning the glass walls.

Since either air or vacuum is used as a medium, there are fewer intrinsic absorption peaks to consider. Operating using engineered absorption peaks from wall resonance rather than intrinsic OH absorption peaks simplifies amplification requirements and provides a larger usable spectrum for Wavelength-Division Multiplexing (WDM). There is also greater flexibility in system design, such as simpler channel planning or a wider range for component selection.

AR-HCF technology, while strong, has had major drawbacks, including difficulties in estimating the effectiveness of various cable designs. Zeisberger and Schmidt established an analytical model for these cables, determining an anti-resonance condition of the fiber walls, predicting leakage-loss scaling, and, overall, a link between geometry and fiber performance [1].

II. METHODS

A. Analytic Model for Complex Effective Index

AR-HCFs are waveguide mediums for light through an air core rather than the standard silica core. This reduces material interactions, which decreases nonlinear interactions. The cores in AR-HCFs have lower refractive indices than the surrounding glass cladding. This makes the guided modes different from traditional step-index fibers. Instead, the guided modes are leaky modes that are dependent on the anti-resonant reflection from the thin glass cladding [1]. The main focus of Zeisberger and Schmidt’s model is to provide an alternative to the simulation design by using an analytic model that parametrizes the fiber geometry to account for attenuation and loss.

The paper starts by defining the complex effective index in the lossy mode [1],

$$n_{\text{eff}} = \text{Re}(n_{\text{eff}}) + j \text{Im}(n_{\text{eff}}) \quad (4)$$

The real part models how the lossy mode propagates through the fiber. The ordered terms account for phase velocity, group velocity, and dispersion. The imaginary term models fourth-order leakage from the core due to material effects. The paper models how the waveguide transports light and how it attenuates along the fiber path.

The effective index is connected to the propagation through [1]:

$$\beta = \frac{2\pi}{\lambda} n_{\text{eff}} \quad (5)$$

β quantifies the phase accumulation of light along the fiber path. Once the effective index is known, the propagation behavior of lossy modes can be modeled to predict dispersion and attenuation. This framework uses geometric parameters to predict key fiber-optic parameters rather than brute-force simulation.

To make the problem geometric, the paper approximates the cylindrical glass cladding as a single planar dielectric film [1]. This is valid only when the radius of the core is much larger than the wavelength of the lossy mode ($R \gg \lambda$). Under this geometric approximation, the authors build a model that reduces the complexity of working with a cylindrical (infinite number of reflection points) system to one that still supports waveguide behavior.

The key finding was the confinement loss for the lossy modes [1]:

$$\alpha \propto \frac{1}{R^4} \quad (6)$$

In traditional fibers, loss is often a third-order term that accounts for material absorption or scattering within the fiber, leading to attenuation. However, the AR-HCF's attenuation is due to leakage through lossy modes. The dependence shows that increasing the radius can significantly reduce signal attenuation.

This paper shows that geometric modeling accurately reproduces the real and imaginary parts of the effective index across transmission bands. This helps predict and analyze resonance positions and dispersion, as validated by simulation [1].

B. Physical and Scaling Constraints

Although the AR-HCFs offer clear benefits over their traditional counterparts in attenuation, their performance is severely limited to anti-resonant guidance. Unlike total internal reflection in traditional fibers, anti-resonant reflection does not perfectly confine propagating signals. This is why the modes are inherently leaky [3]. The paper makes this confinement issue explicit by associating it with the imaginary part of the effective index [1].

Another limiting physical condition for AR-HCFs is the wall thickness relation, which imposes a narrow range for determining whether the glass layer is a strong reflector. A simplified expression shows [1]:

$$t = \frac{m\lambda}{2n_g} \quad (7)$$

t is the thickness of the glass wall, m is the resonance order, λ is the wavelength of the operating mode, and n_g represents the glass refractive index. To fulfill the anti-resonance condition, the fiber must operate in wavelength regimes between these resonance peaks, thereby inducing destructive interference in the glass and causing strong reflection. This shows the importance of selecting the wall thickness. The glass layer supports anti-resonant transmission at operating wavelengths. If the thickness is not carefully selected, the wall

becomes resonant, leading to significant losses. One engineering constraint is manufacturing the AR-HCFs' glass cladding to maintain precise wall-thickness control. Additionally, during fabrication, fiber surface roughness and glass imperfections can shift resonance positions and increase leakage. Therefore, fabricating a fiber whose physical dimensions closely match the ideal geometric design is essential for the anti-resonant mechanism.

This also explains why AR-HCFs operate over select transmission bands rather than broad wavelengths like traditional fibers. The fiber geometry that performs well in one band may not operate anti-resonantly on another. This is why the wavelength is directly related to the fiber geometry through its radius [1]. An important distinction to make is that the fiber is only low-loss when the anti-resonant condition is met.

Another limitation is bending sensitivity. If the fiber is bent, the distributions shift, and radiation grows. This increases the signal's attenuation. This loss is approximated through [4]:

$$\alpha_{\text{bend}} \propto \exp\left(-\frac{R_b}{R_c}\right) \quad (8)$$

where R_c models the characteristic bend radius of the fiber. This shows how confinement decreases under curvature. Even if the fiber is modeled to perform well, realistic deployment would require it to be tolerant of bending in systems.

C. VPI Simulation Setup

For the simulation, an Anti-Resonant Hollow Core Fiber was placed in the same setup as a traditional single-mode fiber and a system with a Dispersion-Compensating Fiber.

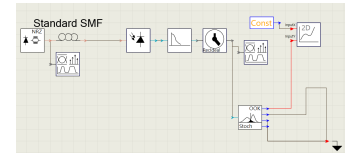


Fig. 1. VPI Photonics Simulation setup

The signal is sent down the fiber, then received by a Photodiode, passed through a low-pass Bessel filter, and graphed by a signal analyzer. The signal is also sent to a BER analyzer to get a BER value. To model an AR-HCF in the software, a Nonlinear Single-Mode Fiber was used, and the parameters were then adjusted.

f	GroupRefractiveIndex	1.003		S
f	Attenuation	0.2e-3	dB/m	S
f	AttFile	Name		S
f	Dispersion	2e-6	s/m ²	S
f	DispersionSlope	0.02e3	s/m ³	S
f	NonlinearIndex	2.2e-22	m ² /W	S
f	CoreArea	160.0e-12	m ²	S

Fig. 2. Anti-Resonant Hollow Core Fiber parameters

The parameters used in the simulation are from the properties of prevalent hollow core fibers being used or tested. The schematic uses a TxExtModLaser to represent an externally modulated laser, the most commonly used type of laser in industry. It uses a PIN photodiode with a responsivity of 1 and zero dark current.

The laser input power, input wavelength, and fiber length were varied separately to see how each fiber type responded. For experiments where the fiber length was kept constant, it was set to 80km, a standard distance between repeater stations for undersea fiber systems. In experiments that kept the signal frequency constant at 193 THz, a common frequency for single-mode fiber communication. In experiments that kept the laser power constant, it was set to 5mW.

III. RESULTS

A. Analytic Model Results

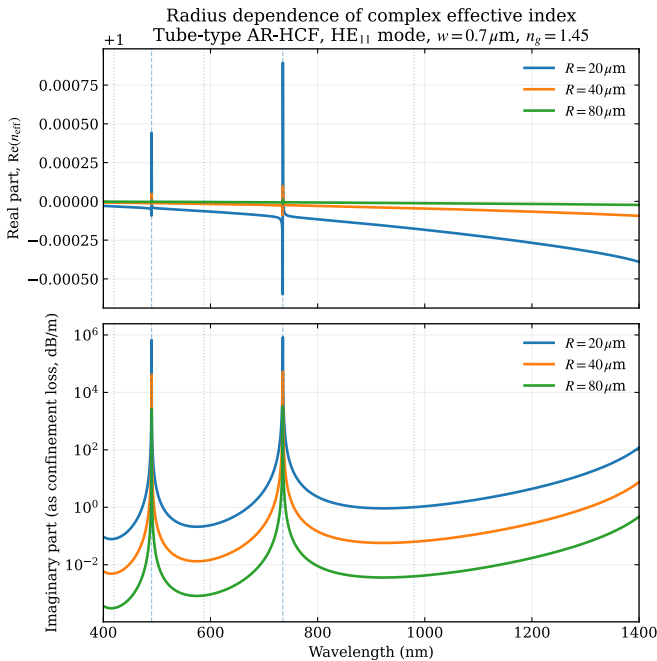


Fig. 3. Comparison of AR-HCF core radius and confinement loss.

B. HCF vs Traditional Fiber: Loss and Dispersion Comparison

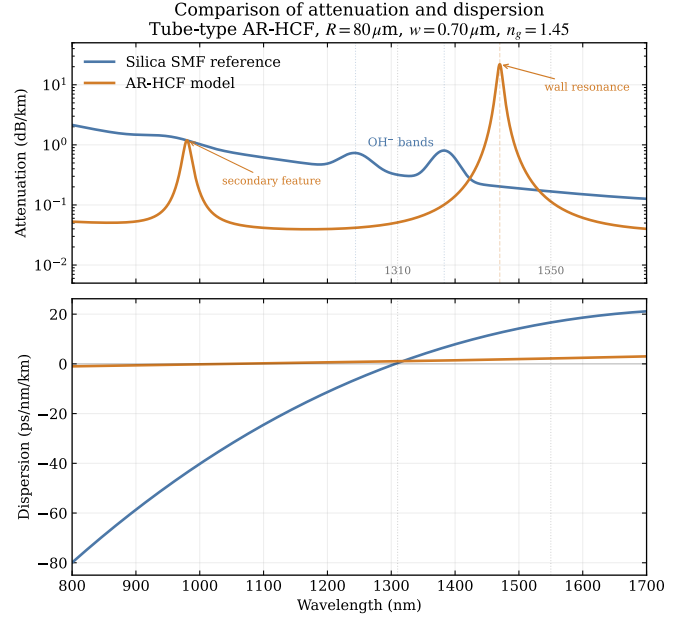


Fig. 4. Attenuation and Dispersion vs. Wavelength of a sample silica SMF and an AR-HCF.

C. VPI Simulation Results

When varying the power from a standard 5mW to 100 mW, the HCF fiber performed much better than the others. It had a BER that was estimated to be zero by VPI, compared to a BER of .019 in the DCF configuration and 3.6×10^{-8} for the standard SMF. This shows the efficiency of AR-HCFs at high power.

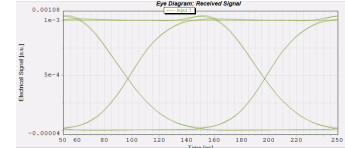


Fig. 5. Hollow Core Fiber Eye diagram at high power

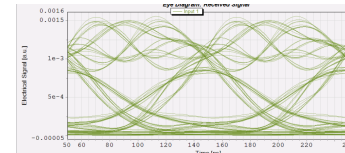


Fig. 6. Standard SMF eye diagram at high power

When the fiber length was increased to 200km, the standard SMF had a BER of 3.2×10^{-5} , whereas the dispersion compensating fiber and hollow core fiber both had BERs on orders less than 10^{-42} . When varying the frequency from 193 THz to 350 THz, the BER of the standard SMF was .2. In contrast, the BER of the HCF was estimated to be zero, showing the performance efficiency and larger range of frequencies available to Hollow Core Fibers.

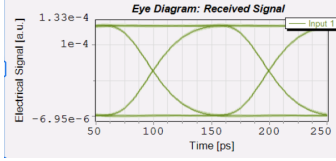


Fig. 7. HCF eye diagram at 350 THz

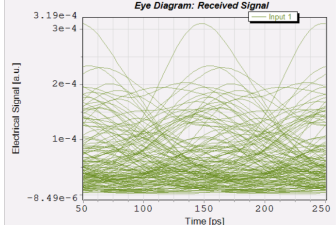


Fig. 8. Standard SMF eye diagram at high power

Our results align with the expectations for Hollow Core Fibers, showing their efficiency and low attenuation over long distances, at high powers, and at high frequencies.

While our results were good, the simulation wasn't ideal, and there were some factors that could have contributed to some inconsistencies, like being given a BER value of exactly zero on some trials. Ideally, the VPI Component Maker software would've been used to make a custom HCF that could be used in simulation. However, we didn't have access to this, so a standard non-linear single-mode fiber with different parameters was used. This led to some issues with the simulation, an error occurred every time the laser power was greater than 100 mW, and the BER for the HCF and also the DCF system was given as exactly zero in some trials. This limited the amount of available data in the simulation, but the data collected still shows that HCFs can perform well at high power and high frequencies.

IV. DISCUSSION

A. Real World Applications of AR-HCFs

An anti-resonant hollow core fiber has several practical applications where its unique properties, over a traditional fiber, provides advantages.

A key important application is in long distance communication systems. Because in a AR-HCF light propagates through air, the effective refractive index is close to one. This thus allows a signal to be able to travel closer to the speed of light, which compared to silica fibers where the propagation is slower. Over long distances, such as undersea communication or large scale data center interconnects. Small reductions in latency add up into a meaningful improvement in performance.

Additionally, in high power laser delivery an AR-HCF can prove useful. Compared to a conventional fiber which is subject to stronger nonlinear effects like Kerr nonlinearity, Raman scattering, and Brillouin scattering. These limit the maximum power that can be transmitted without distortion. Since an AR-HCF confines most of the optical field in its air core, these interactions are significantly reduced, which allows higher optical powers to be transmitted. Because of

this, the fibers are well suited for industrial laser systems, medical procedures, and scientific applications which requires a high intensity beam.

In wavelength areas that silica fibers cannot perform well, an AR-HCF proves useful. Due to the light interacting with the glass material less, absorption losses are reduced which lets transmission in Mid-IR and UV ranges. These are very important for applications in spectroscopy, environmental sensing, and chemical detection where absorption lines are used to identify substances.

Additionally, in gas based sensing and nonlinear optics, the hollow core of the fiber can be filled with gases which let the fiber behave as a waveguide and interaction medium. This enables applications such as gas sensing, frequency conversion, and tunable dispersion systems. In cases like these, fiber geometry and gas composition can be adjusted to tailor the optical response as well.

Finally, AR-HCF's have potential applications in precision timing and metrology. Reduced dispersion and a lower non-linear interaction within the fiber makes the timing accuracy better and the signal stability. In systems such as optical clocks, and synchronization networks, this can prove useful.

While this technology is still being developed for large scale deployment, there is strong potential across communication, sensing, and high power photonics systems, and

B. Effect of the Analytic Model

One of the main contributions of the work that had been done by Zeisberger and Schmidt was the development of the analytical framework that was used for describing the complex effective index of leaky modes in these tube-type anti-resonant hollow-core fibers. [1]. Before this work was done, understanding how these fibers had behaved would often require an extensive simulation and/or repeated parameter sweeps to explore how these different geometric patterns had affected performance. Their analytic model provides a more direct way, rather than relying on simulation, to see how the fiber will behave based on its geometry and estimate.

In this model, the effective index of the guided modes is written in a complex quantity shown by:

$$n_{\text{eff}} = \text{Re}(n_{\text{eff}}) + j \text{Im}(n_{\text{eff}}), \quad (9)$$

The real component describes the optical mode and how it propagates along the fiber and determines properties such as phase, velocity, and dispersion. The imaginary component represents attenuation that is caused here by the leaking of light through the glass surrounding the core. By using a complex index, the model shows how the mode propagates and how it loses power with a mathematical description. This proves very useful.

Another important insight from this paper is that when the core radius is bigger than the wavelength of the guided light, the glass wall surrounding the hollow core can be approximated as a dielectric, planar film. This approximation means that the anti-resonant reflection can be seen as interference within the thin glass layer in the fiber. With this, a clearer physical explanation for why wall thickness, refractive index,

and its contrast, and wavelength are such strong points in whether light remains confined or leaks out of the structure.

The scaling behavior of confinement loss is also important to analyze. The paper shows that loss scales with the inverse of the 4th power of the core radius.

$$\alpha \propto \frac{1}{R^4}, \quad (10)$$

where R is the core radius. This shows that an increase in the core size, even the slightest, can significantly reduce attenuation. This is useful as it can give design intuition to reduce the need for repeated optimization.

Once the effective index is known, the propagation constant can be written as

$$\beta = \frac{2\pi}{\lambda} n_{\text{eff}}, \quad (11)$$

This allows quantities such as phase velocity, attenuation, and dispersion to be seen and analyzed within a single framework.

Overall, the analytic model shows and demonstrates both predictive capability and a clearer understanding of how AR-HCF geometry influences the transmission optically.

C. Wider Effects of AR-HCF Technology

Anti-resonant hollow-core fibers address the limitations of traditional solid-core silica fibers. In conventional fibers, light propagates through silica with a refractive index of $n_{\text{silica}} \approx 1.45$, so the signal moves at about $c/1.45$. In AR-HCFs, the optics stay in an air-filled core where $n \approx 1$, so the signal can travel much closer to the speed of light [1]. Over long distances, that reduction in latency can become meaningful in many applications as it is roughly 30 percent faster.

Additionally, another advantage of AR-HCF's is that it reduces nonlinear optical effects. In a traditional fiber, the optical fields overlap with the silica core. This leads to nonlinear effects like nonlinearity and scattering. These nonlinear interactions distort signals and limit the launch power. On the other hand, in AR-HCFs, only a small portion of the optical power input interferes with the glass structure, which reduces these nonlinear effects. Thus, higher optical power transmission and signal quality in high-bandwidth communication systems are possible.

AR-HCFs also enable transmission through them at wavelength ranges, which silica fibers cannot handle due to higher absorption through their material. Since the field travels mostly through air, the effect of absorption is reduced, which makes it possible to operate in regions such as ultraviolet and mid-IR. These are useful for applications such as spectroscopy, sensing, and high-power laser delivery . [1].

The hollow core structure also has another key advantage in that it can be filled with gases. This allows the fiber to function as an interaction region. These fibers are used in nonlinear optics experiments, dispersion control, and their tunability, and gas-based sensing systems. Because of these properties, AR-HCFs are useful not only for communication systems but also for advanced photonic applications.

D. Practicality and Cost Considerations

Even though AR-HCF Technology has several advantages, theoretically, the practical deployment requires careful considerations in the engineering realm. The analytic model developed by Zeisberger and Schmidt assumes a tube-type anti-resonant geometry where the radius of the core is much larger than the wavelength of the light. Under these conditions, the glass boundary can be approximated. While this approximation captures a lot of the behavior of the fiber, real fabrication imperfections can lead to more loss.

One of the most important challenges is the manufacturing tolerance. Small variations in thickness in the walls can shift the resonance conditions and lead to more loss. How rough the surface is, defects, and structural variations can also introduce scattering that isn't fully captured in the ideal analytical models. Also, AR-HCFs are more sensitive to bending than a conventional fiber, as the bending changes the distribution of the optical modes and increases radiation loss.

Even with these challenges, the model still improves the design for AR-HCF systems. The numerical simulations that can be run for different fiber geometries are time-consuming and computationally expensive. The analytic relationship between geometry and performance given here allows engineers to estimate loss, dispersion, and resonance behavior before running detailed simulations or fabricating devices. This reduces design iterations and lowers development costs.

Although AR-HCFs are still being fixed, refined, and changed for deployment on larger scales, their reduced nonlinear reactions, low latencies, and improved design tools provide a strong potential for future optical communications and sensing technologies.

V. CONCLUSION

Overall, this project looked at the anti-resonant hollow-core fibers as a good alternative to traditional solid-core silica fibers. Additionally, this project also analyzed the analytic model that was developed by Zeisberger and Schmidt for the complex effective index of leaky modes in tube-type AR-HCFs [1]. Conventional fibers guide light through total internal reflection, while, unlike a conventional fiber, an AR-HCF guides light through anti-resonant reflection from a thin glass wall that surrounds an air-filled core.

The guided modes are inherently leaky, and because of this, they are described using a complex effective index. The real component determines how the optical mode propagates and disperses along the fiber. The imaginary component represents confinement loss caused by light leaking through the glass boundary. The analytic model links these two properties directly to geometry, allowing parameters like core radius, wall thickness, and wavelength to be connected to fiber performance.

The strong dependence on loss on the core radius is a significant result, following $1/R^4$ scaling. This provides insight important for design and demonstrates how changes in geometry can reduce attenuation in these fibers by a lot.

Beyond the analytic model, the technology offers several advantages over the traditional silica fibers. Properties that are

hard to achieve in a solid core fiber are more accessible, such as reduced propagation latency, less nonlinear interactions, and access to wavelengths that couldn't be used otherwise. Additionally, the hollow core allows gas-filled operation, which has applications in sensing, tunable photonic systems, and nonlinear optics.

Overall, this work by Zeisberger and Schmidt provides a foundation and practical design guidance for these anti-resonant hollow-core fiber systems. By replacing simulation with an analytic framework, the model provides a key understanding of how the hollow core fiber can support the development of the next generation of communication technologies.

VI. CONTRIBUTIONS

TABLE I
CONTRIBUTIONS OF EACH GROUP MEMBER

Task	Contributors
Abstract	Nicholas
Introduction	Nicholas
Methods	Pranov
Results	Nicholas, Dash
Discussion	Pranov, Gavin
Conclusion	Gavin

REFERENCES

- [1] M. Zeisberger and M. A. Schmidt, "Analytic model for the complex effective index of the leaky modes of tube-type anti-resonant hollow core fibers," *Sci Rep*, vol. 7, no. 1, p. 11761, Sep. 2017.
- [2] O. Humbach, H. Fabian, U. Grzesik, U. Haken, and W. Heitmann, "Analysis of oh absorption bands in synthetic silica," *Journal of Non-Crystalline Solids*, vol. 203, pp. 19–26, 1996, optical and Electrical Properties of Glasses. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0022309396003298>
- [3] J. A. Snyder and J. D. Love, *Optical Waveguide Theory*. London, New York: Chapman and Hall, 1983.
- [4] Y. Zheng, P. Shum, S. Liu, B. Li, Y. Xiang, Y. Luo, Y. Zhang, W. Ni, Z. Wu, X. Q. Dinh, S. Zeng, J.-L. Auguste, and G. Humbert, "Experimental and numerical investigation on hollow core photonic crystal fiber based bending sensor," *Optics Express*, vol. 27, pp. 30 629–30 638, 10 2019.